

THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

PROJECT PROPOSAL

Securing Hospital System Using Role-Based Access Control (RBAC)

Submitted By:

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1. Introduction

Hospitals manage highly sensitive information, including patient medical records, staff details, appointments, and administrative data. If this information is accessed by unauthorized users, it may lead to privacy violations, data exposure, and misuse of confidential records. A secure hospital management system is therefore important for protecting information and ensuring that users can access only the resources required for their duties.

This project proposes a hospital management system secured through Role-Based Access Control (RBAC). Under RBAC, permissions are assigned to predefined roles instead of being assigned separately to every user. The proposed roles may include administrator, doctor, nurse, receptionist, and patient. Each role will have access only to the features and information relevant to its responsibilities.

2. Problem Statement

A hospital system may contain confidential information that should not be visible or editable by every user. Without clearly defined access permissions, users may access restricted pages or records beyond their responsibilities. The project will address this problem by implementing role-based authorization and session-based authentication to restrict access to protected functions and information.

3. Aim and Objectives

The main aim is to design and develop a secure hospital management system that protects sensitive information through Role-Based Access Control.

- To define user roles such as administrator, doctor, nurse, receptionist, and patient.
- To assign permissions according to each role's responsibilities.
- To implement a login system with session-based authentication.
- To restrict access to dashboards, pages, and records according to assigned roles.
- To use PHP for server-side application logic and MySQL for database management.
- To test the system against unauthorized access attempts and verify that role restrictions work correctly.
- To keep the design modular so that additional roles and security features can be added later.

4. Proposed System and Main Features

- **User authentication:** Users will log in with their credentials, and the system will create a session after successful authentication.
- **Role-based dashboards:** Each user will be directed to a dashboard appropriate to their assigned role.
- **Authorization checks:** Protected pages and actions will check the user's role and permissions before allowing access.

- **Controlled record access:** Doctors, nurses, receptionists, patients, and administrators will have different levels of access according to the permissions defined for them.
- **Database management:** MySQL will store user accounts, role information, and relevant hospital-system data.
- **Security testing:** Testing will verify that users cannot access restricted functions simply by entering a protected page's URL directly.

5. Methodology

The project will follow a structured development approach. First, the system requirements and user roles will be identified. Next, the database structure and login interface will be designed. Authentication will validate user credentials and create a session for a valid user. The RBAC authorization layer will then check the user's role and permissions before granting access to dashboards, pages, and database operations. Finally, the system will be tested using accounts with different roles, including attempts to access restricted resources.

1. Requirement analysis and role/permission definition
2. System and database design
3. Implementation of the login and session-management mechanism
4. Implementation of role-based authorization in PHP
5. Integration with the MySQL database
6. Functional and access-control testing
7. Documentation and preparation of the final demonstration

6. Tools and Technologies

- Frontend: HTML, CSS, and JavaScript
- Backend: PHP
- Database: MySQL
- Local development environment: XAMPP or a similar PHP/MySQL environment
- Testing: Manual functional testing with separate user accounts and roles

7. Scope of the Project

The project will focus on access control within a hospital management system. Its scope includes user login, role-specific dashboards, permission checks, and controlled access to relevant hospital records. Advanced capabilities such as audit logging, multi-factor authentication, HTTPS configuration, and detailed healthcare compliance controls may be considered as future enhancements, depending on the available time and project requirements.

8. Expected Outcomes

- A working prototype of a hospital management system with role-based access.
- Separate access permissions for the defined user roles.

- Reduced opportunity for unauthorized users to view restricted pages or records.
- A database-backed application developed with PHP and MySQL.
- Test results demonstrating whether access restrictions behave as intended.

9. Future Enhancements

Future versions may include secure password hashing, audit logs and activity trails, multi-factor authentication, HTTPS-encrypted communication, finer-grained permissions, and improved monitoring of user activity. These enhancements can improve accountability and strengthen the system's security.

10. Conclusion

The proposed project will demonstrate how Role-Based Access Control can be used to secure a hospital management system. By combining session-based authentication with role-based authorization, the system will aim to ensure that users can access only the information and functions appropriate to their responsibilities. PHP and MySQL will provide the main application and database technologies, while testing will evaluate whether the access restrictions work correctly.

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